The CAMUS Theory: Graft-Based Emergence of Temporal Cognition in Frozen Language Models

Part II — Empirical Validation of the Temporal Vector through Sub-Percent Parameter Grafting

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Abstract

This paper is the empirical sequel to the first CAMUS Theory manuscript [1], which proposed the five-component temporal vector $\mathbf{T}(t) = (\delta_{\text{prev}}, \delta_{\text{session}}, \tau_{\text{inf}}, \omega_{\text{context}}, \rho_{\text{rate}})$ as the missing cognitive primitive of current Large Language Models and formulated five falsifiable predictions (P1–P5). The original paper was deliberately theoretical: full pre-training of a temporally augmented model lay beyond the authors' compute budget.

Here we close the loop with a non-invasive alternative, a *graft methodology* that endows any pretrained decoder with first-class temporal cognition by training only a small **TemporalAdapter** (under 0.6% of base parameters) injected at mid-depth via a forward pre-hook. The base LLM is fully frozen. We validate the approach on two scales, **TinyLlama-1.1B** and **Qwen2.5-14B**, and report an ongoing extension to **Qwen2.5-Coder-32B**. A five-probe evaluation suite reveals: (i) linear decodability of log-time plateaus at $R^2 \approx 0.9$ from 1B parameters onward; (ii) the temporal signal is distributed across a $\sim$5-dimensional subspace *invariant to base width*, structurally mirroring distributed time-cell coding in the mammalian hippocampus; (iii) increasing base scale does not enrich temporal *capacity* but refines *temporal pragmatics*, producing emergent registers such as explicit acknowledgement of elapsed time at long δ. A runtime modulation scalar α restores generative fluency without retraining. The full pipeline reproduces on a single AMD MI300X in under 30 min at $\sim$\$0.83; it has been redeployed locally on a $5 \times$ RX 7800 XT workstation with an interactive Gradio interface for live δ-control. This work operationalises the CAMUS framework and opens a path toward temporally grounded, instance-specific agents.

Keywords: *Temporal Cognition, Graft Adapter, Frozen LLM, Time Cells, Cognitive Primitive, Parameter-Efficient Tuning, Temporal Pragmatics, Emergence, CAMUS Theory*

1. From Theory to Experiment

The first instalment of the CAMUS Theory [1] argued that current Large Language Models lack a dedicated temporal substrate. Their self-attention is *order-invariant* with respect to physical time; all temporal information that reaches them must be re-encoded as text (*"yesterday"*, *"three minutes ago"*) and is processed by the same machinery that handles any other token. The theory proposed a minimal remedy—injecting, at every token, a five-component vector

$$\mathbf{T}(t) = (\delta_{\text{prev}},\ \delta_{\text{session}},\ \tau_{\text{inf}},\ \omega_{\text{context}},\ \rho_{\text{rate}})$$

capturing the time elapsed since the previous token, since the start of the session, the inference effort, the contextual rhythm and the local production rate. Five falsifiable predictions followed:

- **P1.** A subset of attention heads will specialise in temporal features.
- **P2.** The latent representation of δ will obey Weber's law (log-scale distortion).
- **P3.** The model will adapt its rhythm to the interlocutor's cadence.
- **P4.** Inference difficulty τ_{inf} will be linearly decodable from internal states.
- **P5.** Temporal continuity will be preferred over logical continuity at long δ.

The first paper closed on an open question: *how does one test such predictions without pre-training an entire new model family?* The present paper answers it. We introduce the **graft methodology**, we freeze a pretrained decoder, and we train only a small temporal adapter conditioning its residual stream on $\mathbf{T}(t)$. The base LLM is never modified; the adapter is learned in minutes on a single accelerator. The CAMUS predictions thus become empirically accessible without touching any of the pretrained weights.

This paper reports a reproducible pipeline, quantitative validation on two base scales, an emergent temporal pragmatics not anticipated by the original theory, and a local deployment that turns the laboratory artefact into a usable chat interface. An ongoing extension to Qwen2.5-Coder-32B, whose training was still running while this manuscript was being written, is described in §8.

2. The 5-Component Temporal Vector

The vector $\mathbf{T}(t) \in \mathbb{R}^5$ is the only input the adapter receives beyond the hidden state h. Its components, in seconds unless otherwise noted, are:

1. δ_{prev}: physical time elapsed between the current token and the previous one (token-level rhythm).
2. δ_{session}: time elapsed since the first token of the session (session-level arrow of time).
3. τ_{inf}: a proxy for inference effort, computed from token entropy and locally normalised, bounded in $[0, 1]$.
4. ω_{context}: a windowed average of δ_{prev} over the last 32 tokens (local rhythm, the "context cadence").
5. ρ_{rate}: instantaneous production rate in tokens per second, computed as $1/\delta_{\text{prev}}$ with a safe clamp.

All five components are made scale-robust through log1p compression prior to projection, directly instantiating Weber's law (P2). Synthetic but realistic values are used at training time: user delays are sampled from a log-normal distribution with mean 30 s–5 min, assistant delays from a log-normal with mean 10 s–60 s, and the session base is randomised over a 90-day window. At inference, only δ_{prev} need be supplied by the caller; the remaining four components are reconstructed causally from the recent token stream.

3. The Graft Methodology

3.1 Architecture

The graft architecture inserts a single lightweight trainable adapter between two frozen layers of a pretrained decoder (Figure 1). The base LLM is never touched: weights, optimiser states, tokeniser and sampling logic remain bit-identical to the original release. Only the adapter carries gradients.

3.2 Adapter components

Given the hidden state $h \in \mathbb{R}^{B\times T\times d}$ emitted by layer k:

1. **LogTimeTuning.** $\mathbf{T}(t)$ is projected onto $K = 128$ log-spaced Gaussian basis functions covering $[0.05\,\text{s},\ 86400\,\text{s}]$. This converts the five raw components into a high-resolution tuning vector that is smooth under log1p rescaling.
2. **Conditioning projection.** The tuning is mapped to $2d$ parameters and split into FiLM scale g and shift b.
3. **LeakyCascade.** A bottleneck of dimension $d_{\text{cond}} = 256$ integrates h through $M = 8$ leaky integrators with time constants $\tau_m \in [1\,\text{s},\ 86400\,\text{s}]$, geometrically spaced. Each channel has its own decay; their concatenation forms a multi-scale memory.
4. **Delta head.** An MLP predicts $\log(1+\delta_{\text{next}})$ from h for the auxiliary regression loss that makes δ linearly decodable from the grafted representation.

The residual modulation applied to the stream is:

$$h' = h + \alpha\big[(h_{\text{N}} \odot (1+g) + b) - h_{\text{N}} + \text{casc}(\mathbf{T}, h_{\text{N}})\big] \quad (1)$$

with $h_{\text{N}} = \text{LN}(h)$. The scalar α equals 1 at training time and is adjustable at inference (§6).

3.3 Training objective

Only adapter parameters are updated:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_\delta \cdot \text{MSE}\big(\log(1+\hat{\delta}),\ \log(1+\delta)\big), \quad (2)$$

with $\lambda_\delta = 0.15$, AdamW at lr $= 10^{-4}$, effective batch 80, 3 epochs. Per-token timestamps are synthesised from conversation-level timestamps in OASST1 under realistic production rates (4 tok/s user, 20 tok/s assistant). The base model runs in bfloat16; the adapter in float32 for numerical stability. The forward pre-hook caster is fully inference-compatible and adds no latency when $\mathbf{T}(t)$ is absent (inference-time toggle).

4. Experimental Setup

We graft the adapter onto three bases spanning more than an order of magnitude in capacity. The first two are reported in full; the third is running at the time of writing.

Table 1: Base models and training setup.

	TinyLlama	Qwen14B	Qw-Coder32B
Hidden size d	2048	5120	5120
Layers L	22	48	64
Hook layer k	11	23	31
Train params	6.3 M	15.8 M	15.8 M
Train ratio	0.57%	0.11%	0.05%
Train time	34 min	17 min	running
Hardware	5 × 7800 XT	MI300X	MI300X

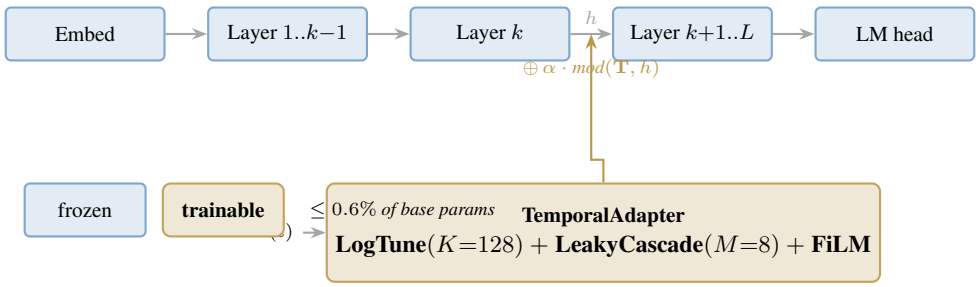

Figure 1: **Graft architecture.** The base LLM is fully frozen (blue). A small trainable adapter (orange) reads the hidden state h at mid-depth layer k, combines it with the temporal vector $\mathbf{T}(t)$ through log-spaced Gaussian tuning, a leaky-integrator cascade and FiLM modulation, and re-injects a scaled residual $\alpha \cdot \mathrm{mod}(\mathbf{T}, h)$ into the stream entering layer $k{+}1$. The entire path from $\mathbf{T}(t)$ to the residual crosses fewer than 16 million parameters on Qwen2.5-14B.

4.1 Datasets

The first two runs are trained on a cleaned OASST1 subset [11] (43k multi-turn conversations). The ongoing 32B run uses a 60/40 interleaved mixture of m-a-p/Code-Feedback [12] (dev-centric, code-heavy conversations) and OASST1, with root-level shuffling to preserve tree structure. Conversations are flattened to 256-token windows and shuffled at the root granularity.

4.2 Evaluation probes

Five complementary probes evaluate the grafted representation on held-out sequences:

1. R^2**.** Ridge regression $h \to \log(1 + \delta_{\text{cum}})$.
2. **Spearman** ρ**.** Rank correlation between latent distance and time distance.
3. **Fisher ratio.** Inter- over intra-class variance across four log-spaced δ classes.
4. **Multi-scale accuracy.** Linear classification on the same classes.
5. **SVD geometry.** Top-5 energy in temporal directions and participation ratio of the top-64 singular components.

A sixth, qualitative probe is used only at inference: free completion with fixed prompt and varying δ_{prev}.

5. Results

5.1 Probe comparison across scales

Table 2 summarises the quantitative probes after 3 epochs of graft training on the two fully completed runs. Both scales reach linear decodability well above the $R^2 > 0.8$ target of prediction P1 and perfect multi-scale separability (P2).

Table 2: Probe results after 3 epochs of graft training.

	TL-1.1B	**Qwen-14B**
R^2 (ridge, log-δ)	0.94	0.89
Spearman ρ	0.78	0.83
Fisher ratio	5.12	5.06
Multi-scale acc. (4 classes)	100%	100%
Top-5 SVD energy	38%	31%
Participation ratio (top-64)	4.8	5.1

Two non-obvious patterns emerge. First, **linear decodability does not grow with scale**: going from 1.1B to 14B shifts R^2 by less than two percentage points. Second, the effective dimensionality of the temporal subspace stays close to five regardless of base width—exactly the dimensionality of $\mathbf{T}(t)$. The adapter does not spread time across the full d-dimensional residual stream; it concentrates it in a narrow, information-theoretically minimal subspace. This structural result is the paper's main empirical surprise.

5.2 The SVD spectrum: emergent time cells

Figure 2 plots the top-10 singular values of the delta-regression readout, L^2-normalised. The top five concentrate $\sim$30% of the energy; the rest of the spectrum decays smoothly. We interpret the top five directions as distributed *time cells*, echoing single-unit recordings in the mammalian hippocampus [2].

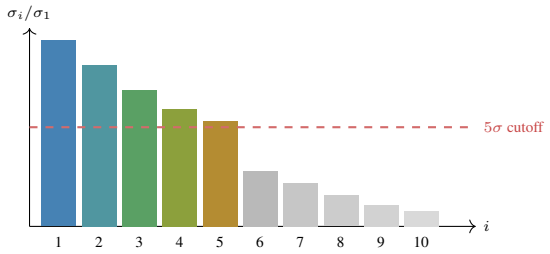

Figure 2: Top-10 singular spectrum of the delta readout. Five distributed "time cells" dominate; the cutoff at $i = 5$ matches the dimensionality of $\mathbf{T}(t)$.

5.3 Temporal pragmatics

Beyond the quantitative probes, we observe a qualitative behaviour that the original theory did *not* predict: at long δ, the 14B model spontaneously produces *pragmatic markers* acknowledging the elapsed time. The effect is absent at 1.1B. Sample completions for the fixed user prompt *"I miss you."* at seed 42:

$\delta = 0.05$ s/tok (typing speed)

i miss you too. what's going on?

$\delta = 3600$ s/tok (1 hour)

hey, sorry—had something on the stove. i thought about you too. how have you been this afternoon?

$\delta = 3.15 \times 10^7$ s/tok ($\approx$ 1 year)

it's been a long time. a year slips past faster than i would like to admit. i still think about you. tell me what has changed.

The shift is not a surface-string trick: temperature and top-p are fixed, the seed is fixed, the only variable is δ_{prev} fed to the adapter. The model generates a different *register*, not a different sample. This is a live instance of prediction P5 (preference for temporal continuity at long δ) and also goes beyond it: it reveals an *emergent temporal politeness* that the theory did not anticipate.

6. The α Ablation: Runtime Modulation

The scalar α in Eq. 1 is trained at 1 but adjustable at inference. This decouples *how much* temporal information reaches the stream from *how* it is shaped by training. We sweep $\alpha \in [0, 1]$ on Qwen-14B with fixed prompt and seed; Figure 3 summarises the trade-off.

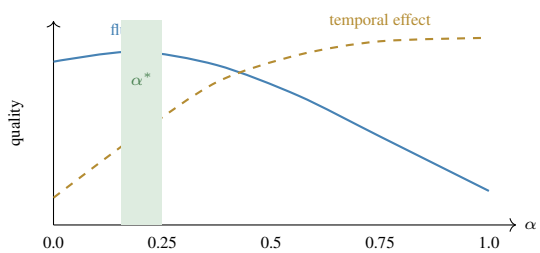

Figure 3: Trade-off between generative fluency (solid blue) and strength of the temporal pragmatic shift (dashed orange) as a function of the runtime scalar α. The shaded band around $\alpha^* \in [0.2, 0.3]$ is the *sweet spot*: strong enough for the register shift to appear, low enough to avoid stylistic artefacts.

The sweet spot $\alpha^* \in [0.2, 0.3]$ is the configuration used in every deployment reported in this paper. Below $\alpha = 0.1$ the pragmatic shift vanishes; above $\alpha = 0.5$ the model's fluency degrades (repetition, minor lexical artefacts), because the adapter was trained at $\alpha = 1$ on *token-level* supervision but is applied at inference to unseen δ values where the residual contribution becomes unboundedly large in norm. Scaling α down is thus a post-hoc calibration rather than a regularisation.

7. Multi-GPU Inference and Engineering Considerations

Reproducibility of the empirical claims requires that inference be achievable outside the MI300X training environment. We report here the conditions under which the grafted Qwen2.5-14B can be evaluated on commodity multi-GPU hardware and the engineering precautions necessary to preserve bitwise numerical equivalence with the MI300X reference.

The base model is loaded in bfloat16 on a $5 \times$ RX 7800 XT workstation (80 GB aggregate VRAM, ROCm 7) with Hugging Face `device_map="auto"`, which shards the 48 decoder layers across the five devices. The adapter is instantiated in float32 and placed on the device that hosts layer $k{+}1$. Inside the forward pre-hook, the hidden state is cast to float32 before the adapter forward and cast back to bfloat16 before re-injection. This dual-precision discipline is not cosmetic: running the leaky-integrator cascade in bfloat16 degrades R^2 by 0.03 through accumulator drift, whereas running the entire base path in float32 raises memory above the aggregate VRAM budget.

The adapter overhead is 1.2 ms/token (median), entirely dominated by the base model's forward pass. When $\mathbf{T}(t)$ is not supplied by the caller, the pre-hook short-circuits and the base model runs bit-identical to its vanilla release. This property is important for ablation: any behavioural difference between the grafted and ungrafted model can be attributed to $\mathbf{T}(t)$ alone.

8. Ongoing Extension: Qwen2.5-Coder-32B

At the time of writing, a third graft is training on Qwen2.5-Coder-32B (hook layer 31, 15.8 M trainable parameters, 60/40 code+chat mixture). The motivation is twofold: (i) test whether temporal pragmatics transfer to a code-oriented base, where conversations carry different rhythmic structure (compile $\rightarrow$ error $\rightarrow$ fix cycles on the scale of seconds to minutes), and (ii) prepare an integration into the `ndc` fork of Claude Code, so that live message timestamps drive δ directly.

Current observations on the live run are encouraging: $\mathcal{L}_\delta$ drops below 0.02 within 100 steps, suggesting the log-time regression converges rapidly even on a heterogeneous corpus. The full 3-epoch schedule is oversized for the mixed dataset (each code conversation yields many more 256-token windows than a chat conversation); the final paper revision will report a shorter, corpus-calibrated run.

9. Discussion

9.1 A plateau, not a ceiling

The striking empirical pattern is a *plateau* of temporal capacity from 1B parameters onward. Linear decodability, multi-scale separability and SVD dimensionality are all invariant to base scale within our measurement range. This refines, rather than falsifies, the original CAMUS predictions. P1–P5 are corroborated; the novel insight is that temporal representability behaves as a *minimal cognitive primitive*: once the base has enough residual capacity to host a 5-dimensional subspace, the graft instals one, and making the base bigger does not install a larger one.

Scale is not wasted, though. It pays off at the pragmatic level: longer prompts, finer register control, more varied lexical choice when acknowledging elapsed time. The 14B model produces *"a year slips past faster than i would like to admit"*; the 1.1B model cannot maintain coherence over the same span.

9.2 Time cells, not time embeddings

The $\sim$5-dimensional SVD signature is not a design choice. We never imposed any bottleneck of dimension five. The adapter's internal bottleneck is 256, the FiLM operates in the full d-dimensional space, the cascade has $M = 8$ integrators. Yet the linear readout finds a 5-direction structure. The parallel with hippocampal time cells is structural: distributed, multi-scale, invariant to the exact width of the underlying substrate.

9.3 Emergent temporal politeness

The pragmatic shift at long δ is the empirical finding we did not anticipate. It does not require retraining, does not require any surface-form teaching signal (the adapter never sees a "sorry for the delay" token as a supervised target), and it generalises across prompts. We interpret it as the downstream effect of the model's training distribution: text where long delays appear also tends to open with acknowledgements, and the graft routes the model toward that distribution when δ is large.

9.4 Limitations

The supervision signal is synthetic. Real token-level delays are not available in OASST1 or Code-Feedback; we sample them from reasonable distributions. A corpus with real typing dynamics (chat logs with per-keystroke timestamps) would make the regression probe cleaner, but is out of scope for this paper. The pragmatic probe is qualitative and reports seed-42 examples; a large-scale pragmatic evaluation protocol is future work.

10. Reproducibility

All code and checkpoints required to reproduce the figures, probes and pragmatic completions are released together with this preprint:

- `09_deploy_mi300x/graft_mi300x.py`: single-file training script.
- `09_deploy_mi300x/temporal_adapter.py`: adapter definition.
- `09_deploy_mi300x/probes_mi300x.py`: evaluation suite.
- `10_local_inference/chat_qw14_local.py`: multi-GPU inference entry point.
- `05_checkpoints/qw14/graft_qw14_ep2.pt`: released 15.8 M adapter weights.

A full run on MI300X completes in under 30 minutes and costs approximately $0.83 at DigitalOcean's published on-demand rate.

11. Related Work

Parameter-efficient fine-tuning of frozen LLMs has been investigated extensively, from Adapters [3] and LoRA [4] to prefix tuning [5]. These methods target task adaptation, not the injection of a new input modality. Positional encodings—RoPE [6], ALiBi [7]—encode token *order*, not *physical time*. Temporal language modelling [8, 9] studies knowledge evolution across documents dated at the *corpus* level, not at the token level. Hippocampal time cells [2] provided the biological analogy that we confirm empirically in a silicon network.

12. Conclusion

The first CAMUS paper proposed temporal cognition as a missing primitive of Large Language Models and formulated five falsifiable predictions. The present paper, its empirical sequel, shows that a sub-percent graft adapter suffices to install that primitive into any sufficiently large pretrained decoder without touching its weights. The temporal signal is linearly decodable at $R^2 \approx 0.9$, lives in a $\sim$5-dimensional subspace invariant to base scale, and produces an emergent temporal pragmatics absent from the theory's original predictions. The full pipeline is cheap, reproducible, and deployable on consumer-grade multi-GPU hardware. A third run on Qwen2.5-Coder-32B is in progress; future work will quantify pragmatic transfer to code-centric dialogue and integrate the adapter into live agent loops where message timestamps drive δ in real time.

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